The Philanthropic Consequence of Government Grants to Nonprofit Organizations

A META-ANALYSIS

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Do government grants displace or leverage private donations to nonprofit organizations? Although research on this topic is flourishing, the findings remain extremely contradictory, creating difficulty in developing a cumulative knowledge available to scholars and practitioners. This study employs a meta-analysis to systematically synthesize the competing findings from the existing literature. Using a sample of sixty original studies with 637 effect sizes, this study finds government grants have almost no correlation with private donations. In addition, this study demonstrates, through meta-regression, that nonprofit subsector variation, organizational age as a control, longitudinal data structure, and endogeneity correction help explain the effect size heterogeneity within and across original studies. Nonprofits should be more concerned about the capacity of competing for different funding sources rather than the tradeoff among them.

Keywords: *government grants, private donations, crowding-in, crowding-out*

A REMARKABLE FEATURE OF CONTEMPORARY government–nonprofit relationships is the close intertwine of the two sectors through networks and partnerships in service delivery and policy implementation (Gazley and Brudney 2007; Smith and Lipsky 1993; Young 2000). Accordingly, government funding represents one important funding source for the nonprofit sector (Lu 2015; McKeever and Pettijohn 2014). The dependence on government funding

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naturally leads to a concern for whether government funding would compromise nonprofit missions and autonomy. Indeed, as resource dependence theory points out, when an organization becomes dependent on external resources, it inevitably becomes subject to external control and loss of independence (Pfeffer and Salancik 1978). For these reasons, nonprofit management scholarship has paid considerable attention to nonprofit–government interactions, examining the impacts of government funding on nonprofit behaviors, such as organizational management, governance, and civic engagement (Chavesc, Stephens, and Galaskiewicz 2004; Mosley 2012; O'Regan and Sharon 2002; Smith 2010).

One strand of research in this regard explores the influence of government grants on private donations within nonprofits, specifically, whether government grants would displace private donations (crowding-out effect) or leverage them (crowding-in effect). This research question has received substantial scholarly attention from a variety of disciplines, such as public administration, nonprofit management, and public finance. There is empirical evidence supporting both the crowding-in and crowding-out lines of argument (Brooks 2000a; Tinkelman 2010). In addition to holding scholarly interest, this question has strong managerial relevance to nonprofit organizations. In particular, nonprofit managers seek to understand the interactions among their revenue streams and design effective financial strategies. Unfortunately, the diverse and competing findings from the existing literature constitute a barrier to offering a clear, consistent knowledge for both research and practice.

This study implements a meta-analysis to aggregate previous quantitative studies on the effect of government grants on private donations within nonprofits. It contributes to the literature by providing a comprehensive synthesis of the existing studies and by exploring the reasons underlying the competing results from those studies. Based on a sample of sixty original studies including 637 effect sizes, this meta-analysis finds only a slight positive correlation (roughly ranging from .01 to .06) between government grants and private donations. Given the negligible magnitude of the association, it may be concluded that government grants have almost no impact on either crowding-in or crowding-out private funding. Put another way, the debate about crowding-in or crowding-out might be unfounded. Moreover, this study finds, through meta-regression models, that nonprofit subsector variation, organizational age as a control, longitudinal data structure, and endogeneity correction help explain the effect size heterogeneity within and across original studies. Nonprofits should be more concerned about the capacity of competing for different funding sources than the tradeoff among them.

The sections are arranged as follows. The next section reviews relevant existing literature and outlines the mechanisms through which government grants may displace or leverage private contributions. The third section introduces the meta-analysis technique and then describes the literature search procedure used to locate original studies. Details on estimating average effect size and identifying moderator variables that explain the variability in effect sizes are presented in the fourth section. The article concludes with a discussion of the managerial implications of the findings as well as some areas for future studies.

Literature Review

Scholarly discussions on the impact of government grants on private giving largely fall into two categories: sectoral and institutional (Borgonovi and O'Hare 2004). At the sectoral level, studies explore potential associations between government expenditure and private donations

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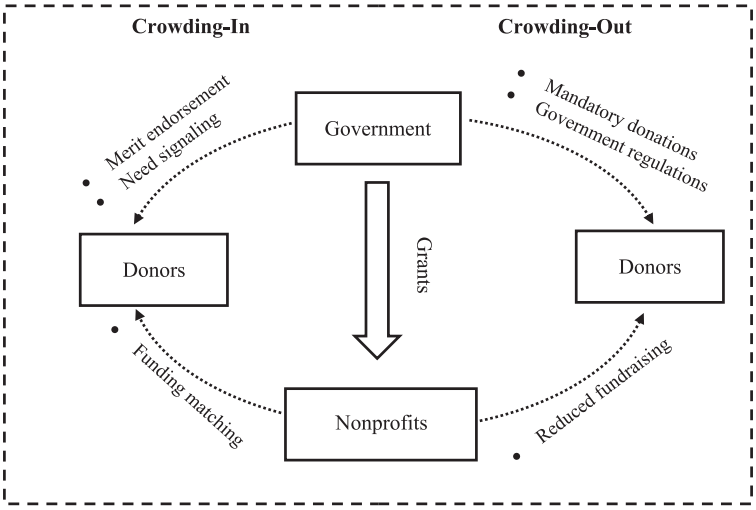
Figure 1. Crowding-Out and Crowding-In between Government Grants and Private Donations



to the entire nonprofit sector or a certain subsector (Brooks 2000a; Schiff 1985; Steinberg 1987). At the institutional level, researchers are interested in the impact of government supports on private donations to individual nonprofits (Andreoni and Payne 2003; Heutel 2014; Okten and Weisbrod 2000). The present study is based on the institutional level and uses individual nonprofits as the unit of analysis. Generally, the interactions between government grants and private contributions in nonprofits may proceed in two directions: crowding-out and crowding-in, with the net effect dependent on the strategic responses of both donors and charities (Tinkelman 2010). Figure 1 summarizes these two routes.

Crowding-Out

The classic crowding-out theory assumes that donors are highly altruistic and care about the total provision of public goods. Based on this assumption, donors as taxpayers would consider government funding to be a substitute for voluntary donations, as taxation conveys their contributions to recipients. When a nonprofit receives more supports from government, donors would consciously reduce giving to this organization. In this way, there is a complete dollar-for-dollar replacement of private giving by government funding (Roberts 1984; Warr 1982). Andreoni (1989) challenged this classic thinking, arguing that donors are also motivated by the utility from the act of giving to help others (a "warm glow"). According to this perspective, government funding and private giving are not completely substituted, resulting in only a partial crowding-out. In addition, government funding always comes with regulations and requirements on nonprofit operation. Potential mission drift and weakened donor control would thereby lower donation motives (Brooks 2004). However, all these explanations are built on the assumption that donors possess perfect information about the nonprofits they donate to, which is not always the case (Krasteva and Yildirim 2013). Horne, Johnson, and Van Slyke (2005) demonstrated that individual donors do not necessarily have complete information on the financial structures of their beneficiary organizations and link donation decisions to the levels of government funding.

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From a beneficiary organization point of view, nonprofits' responses to government funding would also influence their solicitation of private donations. Nonprofits are generally believed to be social mission–driven actors, with weaker incentives to sedulously maximize their revenue. As a result, an increase in revenue from one source is likely to be associated with a decrease in another source (Okten and Weisbrod 2000). Government funding would thus lower recipients' financial needs and reduce their fundraising efforts. Both Andreoni and Payne (2003) and Hughes, Luksetich, and Rooney (2014) supported this line of argument: government funding leads to significant reductions in nonprofit fundraising expenditures. In sum, quite a few studies fall into this crowding-out group (for example, Andreoni and Payne 2011b; Dokko 2009; Payne 1998; Ribar and Wilhelm 2002; Tinkelman 1999).

Crowding-In

At the same time, government grants may stimulate private donations through three mechanisms: merit endorsement, social need signaling, and matching requirements (Brooks 2004; Rose-Ackerman 1981). First, when donors do not have complete knowledge concerning beneficiary nonprofits and their programs, government funding serves as an endorsement of quality and reliability (Payne 2001). Nonprofits receiving government grants have to go through competitive merit-based selection processes, with extensive review and monitoring. Therefore, uninformed donors can perceive the receipt of government grants as an indicator of trustworthiness and competence. Second, government grants can be considered as a signal of social needs, bringing attention to potential donors and further facilitating the leveraging effect (Brooks 1999). Third, government grants sometimes come with matching requirements that demand that nonprofits raise corresponding money from private sources. This induces nonprofits to increase fundraising efforts to bring in more private donations. Karlan and List's (2007) and Meier's (2007) field experiments confirmed that matching mechanisms increase solicitation response and charitable revenue. In short, a number of studies support this crowding-in argument (for example, Borgonovi and O'Hare 2004; Herzer and Nunnenkamp 2013; Heutel 2014; Khanna and Sandler 2000; Okten and Weisbrod 2000).

Other studies, in addition to those making traditional crowding-out and crowding-in arguments, contribute to our understanding of the complexity of government–private funding interaction. Some have argued that the impact of government funding on private donations is overstated. Brooks (1999), for example, tested the causality between government subsidies and private donations in symphony orchestras and concluded that these two funding sources are independent. Some have noted that the relationships between government grants and private contributions are not necessarily linear. Nikolova (2015) examined government–nonprofit interaction using international development nonprofits. She showed that when government contribution is below one-third of organizational revenue, it attracts private philanthropy. However, when government support exceeds that point, it displaces donations.

In sum, all these competing arguments and bodies of evidence, though they deepen our understanding of the complicated interaction between government grants and private contributions, also pose a wide intellectual gap for us to bridge. The need here is "not additional empirical data but some means of making sense of the vast amounts of data that have been accumulated" (Schmidt & Hunter 2015, 17). The development of a coherent knowledge on this topic calls for a systematic synthesis.

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Methodology

This study used a meta-analysis to aggregate the existing literature on government grant–private donation interaction. Meta-analysis, as Glass (1976, 3) termed, is a "statistical analysis of a large collection of analysis results from individual studies for the purpose of integrating the findings." In essence, meta-analysis is a literature review technique. However, unlike conventional narrative literature reviews, a meta-analysis applies statistical methods and criteria to summarize and evaluate quantitative studies and thus enables the extraction of generalizable conclusions from contradictory studies (Stanley 2001). Recent advancements in meta-regression techniques further help examine the factors that lead to different conclusions in existing studies (Thompson and Higgins 2002). The meta-analysis in this study generally follows the procedures suggested by Ringquist (2013).

As meta-analysis aims for aggregation of effect sizes within and across studies, the sample for the study depends on a literature search to identify original studies, both published and unpublished. The literature search in this study employed two complementary search strategies: a computerized search and a manual search. The computerized search was conducted to locate studies using electronic databases in three categories: academic search engines (EBSCO, JSTOR, ERIC, ISI Web of Knowledge, Worldcat), working papers (SSRN, NBER, EconLit), and PhD dissertations (ProQuest). Each of these sources was searched in a full-text format using the following keywords: *crowding out*, *crowding in*, *government*, *donation*, and *giving*. After collecting an initial list of studies, the manual search was conducted using Google Scholar to trace backward all references cited in the articles and trace forward all articles that cited the original articles. The entire literature search was completed in late February 2015.

Original studies were included in the meta-analysis if they met the following criteria: (1) the study applied quantitative methods and provided sufficient statistical information for the calculation of effect size of the relationship between government grants and private contributions; (2) the unit of analysis was an individual nonprofit, rather than the nonprofit sector, a subsector, or an individual donor; and (3) the dependent variable *private contributions* was operationalized as the amount of money donations from individuals, corporations, and foundations, while the independent variable *government grants* was measured as the monetary amount of grants from different levels of governments.[1] Finally, these search strategies yielded a sample of sixty primary studies, consisting of forty-nine published articles and eleven unpublished articles.

Results

This section first estimates the average effect size between government grants and nonprofit donations from all the original studies, and then explores the moderators that explain the effect size heterogeneity within and across those studies.

Effect Size Analysis

Effect size represents a standardized measure of association between the focal predictor *government grants* and the dependent variable *private donations* and is the fundamental metric and unit of analysis in meta-analysis. An r-based effect size, Fisher's z, was used in this study given that both the focal predictor and the dependent variable are measured continuously. Regression coefficients and other measures of association were first transformed into r-based measures to

allow for comparison.[2] Small sample biases were then corrected by transforming the *r*-based measures into Fisher's *z*. When an original study reported multiple effect sizes, largely resulting from different model specifications or sample restrictions, all relevant effect sizes were included in the meta-analysis to retain the within-study variation (Ringquist 2013). Finally, a total of 637 effect sizes were calculated from sixty acceptable studies (Table 1). These effect sizes ranged from – .8219 to 1.3437, with an unweighted mean of .0266 and a standard deviation of .1488. Table 1 also reports the average effect size for each original study.

Effect size heterogeneity was first assessed before synthesizing these effect sizes to choose between a fixed-effect and a random-effect model. Specifically, the I^2 statistic of 97.6 percent ($p < .0001$) indicates that roughly 98 percent of the variance in effect sizes could not be explained by sampling error, but by high heterogeneity in the effect size within and across studies (Higgins et al. 2003). Given the presence of true effect size heterogeneity, a random-effect specification was employed to calculate the average effect size. The weighted average effect size is .022 ($z = 5.94$, $p < .0001$), with a 95 percent confidence interval of .015 to .029. This average effect size can be interpreted as a positive partial point biserial correlation between government grants and private donations (that is, a crowding-in effect), and the magnitude of the effect is fairly small.

Moderator Analysis

As noted previously, there is significant variation in effect sizes between and within studies. Thus, in addition to estimating average effect sizes, an equally interesting question is why these original studies reach different conclusions concerning government grants and private donations interaction. Toward this end, meta-regression analysis was performed to identify potential moderators that could explain the systematic variability in effect sizes both within and across studies (Stanley and Jarrell 1989).

The meta-regression model in this study was estimated using all 637 effect sizes from sixty existing studies. The dependent variable is the unweighted effect size of the association between government grants and private donations from original studies (transformed into Fisher's *z*). The independent variables, or moderators, are influencing factors on the expected value of the effect size. Specifically, two categories of moderators were identified to predict the dependent variable: (1) characteristics of nonprofits under study, such as subsector and country, and (2) characteristics of research design, such as data structure and control for endogeneity. Again, a random-effect meta-regression model was employed as an extension of a random-effect meta-analysis model that includes effect size-level covariates. The full meta-regression model is specified as follows:

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + b_3X_{3i} + \ldots + b_{13}X_{13i} + b_{14}X_{14i} + e_i$$

Where

Y_i: unweighted effect size in original study *i*

X_1: whether the effect size is estimated from a model using only arts nonprofits, yes = 1 and no = 0

X_2: whether the effect size is estimated from a model using only education and research nonprofits, yes = 1 and no = 0

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Table 1. Effect Sizes at Study Level ($N = 60$)

#	Study	# of Effect Size	Average Effect Size	95% Confidence Interval	% Weight
1	Andreoni & Payne (2011a)	14	−.01	(−.02, −.00)	2.75
2	Andreoni & Payne (2011b)	10	−.01	(−.02, .00)	2.73
3	Balsam & Harris (2014)	15	.01	(−.02, .05)	2.30
4	Borgonovi (2006)	5	.15	(.04, .25)	1.04
5	Boumahdi, Carayol, & Llerena (2008)	4	−.09	(−.17, −.01)	1.48
6	Breman (2006)	12	−.06	(−.13, .01)	1.58
7	Brooks (1999)	4	.03	(−.23, .29)	.27
8	Brooks (2000b)	2	.92	(.57, 1.27)	.16
9	Brooks (2003)	3	.35	(.12, .57)	.36
10	Calabrese (2011)	18	.00	(−.01, .01)	2.71
11	Calabrese & Grizzle (2012)	24	−.00	(−.01, .01)	2.72
12	Callen (1994)	3	−.01	(−.24, .23)	.32
13	Chen (2005)	14	.06	(.02, .11)	2.15
14	Chen (2009)	2	.16	(.09, .23)	1.60
15	Cheslock & Gianneschi (2008)	3	.10	(.03, .17)	1.69
16	Colon-Mollfulleda (2008)	2	.14	(.00, .27)	.81
17	Dokko (2009)	3	−.01	(−.03, .00)	2.64
18	Frumkin & Kim (2001)	6	.04	(−.01, .08)	2.23
19	Harris & Ruth (2015)	6	.02	(.00, .04)	2.66
20	Herzer & Nunnenkamp (2013)	8	.21	(.15, .26)	1.90
21	Heutel (2014)	19	.00	(−.00, .01)	2.76
22	Horne (2005)	68	.02	(−.04, .08)	1.77
23	Hughes & Luksetich (1999)	6	.00	(−.15, .15)	.68
24	Hughes, Luksetich, & Rooney (2014)	6	.30	(.18, .42)	.89
25	Jacobs & Marudas (2006)	1	−.20	(−.32, −.07)	.85
26	Jacobs & Marudas (2009)	3	.09	(.06, .11)	2.52
27	Khanna, Posnett, & Sandler (1995)	9	.04	(−.04, .13)	1.38
28	Khanna & Sandler (2000)	8	.08	(−.01, .17)	1.32
29	Kitching (2009)	6	−.01	(−.06, .05)	1.98
30	Marcuello & Salas (2000)	8	.01	(−.27, .30)	.23
31	Marcuello & Salas (2001)	8	.03	(−.26, .31)	.23
32	Marudas (2001)	8	−.04	(−.11, .02)	1.80
33	Marudas (2004)	12	.04	(−.03, .10)	1.75
34	Marudas, Hahn, & Jacobs (2012)	9	−.06	(−.37, .25)	.20
35	Marudas & Jacobs (2004)	6	.03	(−.05, .11)	1.42
36	McCleary & Barro (2008)	3	.04	(−.03, .06)	2.58
37	Monti (2010)	11	−.00	(−.03, .02)	2.55
38	Nicholson-Crotty (2011)	10	.06	(.03, .08)	2.57

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Table 1. (Continued)

#	Study	# of Effect Size	Average Effect Size	95% Confidence Interval	% Weight
39	Nikolova (2015)	20	.04	(−.00, .08)	2.29
40	Nunnenkamp & Öhler (2012)	1	.18	(.09, .26)	1.36
41	Okten & Weisbrod (2000)	10	.07	(−.01, .15)	1.42
42	Omura (2010)	32	.08	(−.19, .36)	.25
43	Owalla (2008)	14	.01	(−.03, .06)	2.14
44	Payne (1998)	4	−.03	(−.07, .00)	2.36
45	Payne (2001)	23	.12	(.08, .17)	2.21
46	Posnett & Sandler (1989)	11	−.00	(−.20, .20)	.42
47	Ribar & Wilhelm (2002)	8	−.04	(−.11, .03)	1.62
48	Sav (2012)	10	.07	(−.08, .22)	.68
49	Seaman (1980)	8	.05	(−.24, .35)	.21
50	Sieg & Zhang (2012)	19	.02	(.01, .04)	2.68
51	Smith (2003)	4	.24	(.16, .31)	1.56
52	Smith (2007)	8	.06	(.05, .08)	2.69
53	Tinkelman (1999)	6	−.24	(−.27, −.21)	2.44
54	Tinkelman & Mankaney (2007)	4	−.02	(−.03, −.01)	2.74
55	Tinkelman & Neely (2011)	7	−.01	(−.05, .03)	2.29
56	Wiepking & van Leeuwen (2013)	1	.38	(.13, .63)	.29
57	Wilsker (2011)	25	−.06	(−.07, −.05)	2.75
58	Yetman & Yetman (2003)	24	−.12	(−.26, .01)	.79
59	Yetman & Yetman (2013a)	27	.03	(−.00, .06)	2.47
60	Yetman & Yetman (2013b)	12	.04	(.04, .05)	2.75
	Overall (Study-Level, Random Effects)	637	.03	(.01, .04)	100.00

Note: (1) I-square = 95.2%, p = .000; (2) weights are from random effects analysis.

X_3: whether the effect size is estimated from a model using only environment nonprofits, yes = 1 and no = 0

X_4: whether the effect size is estimated from a model using only health care nonprofits, yes = 1 and no = 0

X_5: whether the effect size is estimated from a model using only human service nonprofits, yes = 1 and no = 0

X_6: whether the effect size is estimated from a model using only international development nonprofits, yes = 1 and no = 0

X_7: whether the effect size is estimated from a model using only public benefit nonprofits, yes = 1 and no =0

X_8: whether the effect size is estimated from a model using US-based nonprofits, yes = 1 and no = 0

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X_9: whether the effect size is estimated from a model including organizational age as a control, yes = 1 and no = 0

X_{10}: whether the effect size is estimated from a model including organizational size as a control, yes = 1 and no = 0

X_{11}: whether the effect size is estimated from a model using cross-sectional or longitudinal data, longitudinal data = 1 and cross-sectional = 0

X_{12}: whether the effect size is estimated from a model including lagged government grants as an independent variable, yes = 1 and no = 0

X_{13}: whether the effect size is estimated from a model including a control for endogeneity, yes = 1 and no = 0

X_{14}: whether the effect size is estimated from a peer-reviewed published literature, yes = 1 and no = 0

Methodologically, two issues complicate the estimation of the meta-regression model: heteroskedasticity and non-independent observations (Ringquist 2013). The concern for heteroskedasticity arises because the effect sizes are calculated from original studies with varying sample sizes. As sample size increases, variability of effect sizes decreases because of reduced sampling error. In addition, the effect sizes calculated from the original studies are not completely independent, largely because most studies generally produce multiple effect sizes per study. Rather than averaging these effect sizes to achieve a single study-level effect size, multiple effect sizes were retained to capture the within-study variation (Ringquist 2013; Thompson and Higgins 2002). As a result, effect sizes from original studies might be correlated because of similarity in data sources and estimation procedures.

This study employed two regression techniques to jointly solve these two challenges. First, a weighted least squares approach handled the heteroscedasticity problem by weighting each estimate by the inverse of the sample size. Based on this, clustered robust standard errors (CRVE) were used to further address the dependence of effect sizes in meta-regression. Viewing multiple effect sizes clustered within original studies, the CRVE model adds to the weighted least squares model with clustered standard errors and weighs all effect sizes equally (Sterne et al. 2002). Second, generalized estimating equations (GEE) were also employed to correct effect size dependence. Unlike CRVE, GEE assumes that effect sizes from the same original study produce less information than those from different studies, and thus adjusts the model estimation by placing less emphasis on effect sizes from studies with multiple effect sizes. Indeed, CRVE and GEE each have their own merits and limitations in correcting the previously mentioned methodological problems (Burton, Gurrin, and Sly 1998; Zorn 2006). Combining these two models allowed conclusions based on the findings to be made with greater confidence (Ringquist 2013). Table 2 presents the results of the meta-regression using the two specification methods. Overall, both the CRVE and GEE models were statistically significant and account for a portion of the systematic variation in effect sizes.

Effect of Nonprofit Subsector on Effect Sizes

Organizations in the nonprofit sector are extremely diverse, with nonprofits in different subsectors displaying dissimilar objectives and operating logics (Steinberg 1986). The meta-regression includes moderators on each nonprofit subsector to examine this line of reasoning.

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Table 2. Meta-regression of the Impact of Government Grants on Private Donations in Nonprofit Organizations ($N = 637$)

Moderator	*CRVE*	*GEE*
Arts	.0920**	.0960***
	(.0378)	(.0374)
Education and research	.0111	.0107
	(.0319)	(.0310)
Environment	−.0054	.0011
	(.0272)	(.0284)
Health care	.0217	.0424**
	(.0170)	(.0176)
Human services	−.0648*	−.0472
	(.0339)	(.0300)
International development	.0121	.0066
	(.0221)	(.0250)
Public benefit	.0338	.0253
	(.0336)	(.0292)
US nonprofits	−.0348	−.0234
	(.0288)	(.0301)
Age	−.0424*	−.0427*
	(.0233)	(.0230)
Size	−.0077	−.0033
	(.0263)	(.0300)
Longitudinal data	.0013*	.0012*
	(.0007)	(.0006)
Lagged grants	.0010	.0138
	(.0231)	(.0324)
Endogeneity	−.0384**	−.0383***
	(.0156)	(.0154)
Peer-reviewed	.0231	.0468*
	(.0145)	(.0238)
Constant	.0635*	.0428***
(Baseline effect size)	(.0374)	(.0173)
F	2.37***	
Wald χ^2		23.13**
Adjusted R^2	.1342	

Note: Robust standard errors in parentheses;
**significant at .1; **significant at .05; ***significant at .01.*

All the nonprofits in our sample were roughly categorized into seven subsectors when possible: arts, education and research, environment, health care, human services, international development, and public benefit.[3] The meta-regression models provide some supports for the subsector difference. The average correlation between government and private funding is

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significantly smaller in models only using human service nonprofits than in those using other nonprofits by approximately .06 ($p < .1$). Again, effect sizes from arts and health care nonprofits are significantly larger than those from other nonprofit samples, with on average differences of .09 ($p < .05$) and .04 ($p < .05$), respectively. In contrast, models employing data from nonprofits in education and research, environment, international development, and public benefit do not show significant differences in average effect sizes. Overall, these findings do point to the moderator role of the nonprofit subsector and are an indication that future studies need to take subsector differences into consideration.

Effect of Country on Effect Sizes

Sokolowski (2013) claimed that country-specific socioeconomic factors would moderate the transaction costs between nonprofit revenue sources, leading to different institutional settings that shape the government grant–private donation interaction. However, the findings from Posnett and Sandler's (1989) study of UK charities and Andreoni and Payne's (2011a) study of Canadian charities showed general consistency with US data. To check potential of cross-country differences, another moderator, country of nonprofits, was added to compare the average effect sizes from US and non-US organizations. In meta-regression models, there is little support for the country-specific argument. Both regression models show no significant difference between the two groups ($p > .1$), even though the parameters imply that US-based samples on average produce smaller effect sizes. This finding illustrates that the impact of government grants on private contributions may be generalizable across countries.

Effect of Organizational Characteristic on Effect Sizes

Two organization-level moderators were included to check whether they help account for the variation in effect sizes within and across studies. First, organizational age may moderate the interaction of public and private funding in that government grants have a greater impact on young nonprofits than on established ones (Heutel 2014). When potential donors have limited information about young organizations, government funding acts as a signal of quality and reputation, effectively mitigating information asymmetry and leveraging private contributions. However, as nonprofits grow, this signaling effect decreases and the crowding-in effect would ultimately disappear. The meta-regression analysis supports this line of argument. The effect sizes estimated from models controlling for age are on average significantly smaller than those from models without such a control by approximately .04 ($p < .1$). This finding emphasizes the necessity of including some measures of organizational age in future studies.

Second, organizational size may also moderate public–private funding interaction. In addition to the signaling effect due to imperfect information, being awarded a grant has a bigger impact on smaller nonprofits because these organizations do not possess commensurate administrative and fundraising capacities to effectively handle different revenue sources simultaneously (Andreoni, Payne, and Smith 2014). Hughes et al. (2014) showed that government supports exert a positive impact on private supports in larger orchestras but a negative one on smaller ones. However, the meta-regression models do not clearly underscore the moderator role of organizational size. Although the average effect size estimated from models controlling for size seems smaller, the difference is not statistically significant at .1 level in both models. Future studies may continue exploring the moderating role of organizational size.

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Effect of Research Design on Effect Sizes

Three moderators related to research design were added to the meta-regression to explore whether these design components significantly improve study quality. First, the influence of data structure was examined. It has been widely acknowledged that longitudinal data are superior to cross-sectional data because they enable the researcher to track the dynamics of public–private funding interaction over time. The results from the meta-regression analysis endorse this consensus. The average effect size estimated from longitudinal data is slightly larger than that from cross-sectional data by .001 ($p < .1$). In sum, data structure is not independent from the estimate of effect sizes in original studies, with longitudinal studies yielding larger impacts.

Second, the influence of government grants may be cumulative over time and long lasting (Andreoni et al. 2014). Organizational and donor responses to nonprofits' receipt of government grants are likely not to be contemporaneous; they can take some time, sometimes with a delay of more than a year. A common methodological strategy to address this issue is to introduce lagged government grants as an independent variable in regression models. Indeed, the parameters estimated in both meta-regression models indicate a small insignificant difference between effect sizes calculated from models based on the current level of government grants and those from models including lagged government grants ($p > .1$). Thus, this design component fails to explain the systematic variation of effect sizes within and between studies.

Third, the exploration of government grants and private donations in nonprofits has to address the concern for endogeneity, that is, government funding and private donations may be jointly determined (Andreoni and Payne 2003; Knight 2002; Ribar and Wilhelm 2002). For example, nonprofits with effective fundraising may solicit more funding from both sources. Without any statistical control for the endogeneity problem (such as fixed effects and instrumental variables), the estimated relationships could be biased. Thus, a moderator was included in the meta-regression to test this argument. The results from both models underscore the imperative of endogeneity correction: the average effect size is markedly smaller from models embracing some endogeneity-corrected estimates by about .04 ($p < .05$). In future studies, the mere use of regular regressions without any adaptation of endogeneity may overstate the impact of government grants.

Effect of Publication Bias on Effect Sizes

One issue that disturbs the validity of meta-analysis is publication bias: if there are studies that appear to be systematically excluded from our sample, the average effect size estimated could be inaccurate. Unfortunately, the methods to address publication bias in meta-analysis are not sufficiently advanced. This study performed three types of diagnostics to check the potential of publication bias. First, as a graphic illustration, a funnel plot is presented in Figure 2 for the 637 effect sizes. The plot is slightly asymmetric, with a few more positive effect sizes on the right. Second, Begg and Eggers tests were conducted. Although the Begg test ($Z = .52$, $p > .1$) shows no publication bias, the Egger test raises some concern ($t = 1.70$, $p < .1$). Third, to further compare published and unpublished studies, a moderator on whether a study comes from a peer-reviewed published outlet was incorporated in the meta-regression. Only the GEE model shows a significant difference of .05 ($p < .1$) in effect sizes between published and unpublished studies. In sum, although there may exist a systematic difference, the impact of the difference is fairly small. Thus, the analysis here would not be seriously contaminated by publication bias.

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Estimated Effect Sizes under Various Scenarios

Based upon the moderator analysis presented here, I estimated in Table 3 the average effect sizes (Fischer's *z*) under different scenarios from both the CRVE and the GEE models. The first row reports average effect sizes under the meta-regression baseline scenario, when all the moderators in the regression model take on values of 0. The second through ninth rows report the average effect sizes in different nonprofit subsectors from meta-regression models under the baseline

Figure 2. Funnel Plot of 637 Effect Sizes

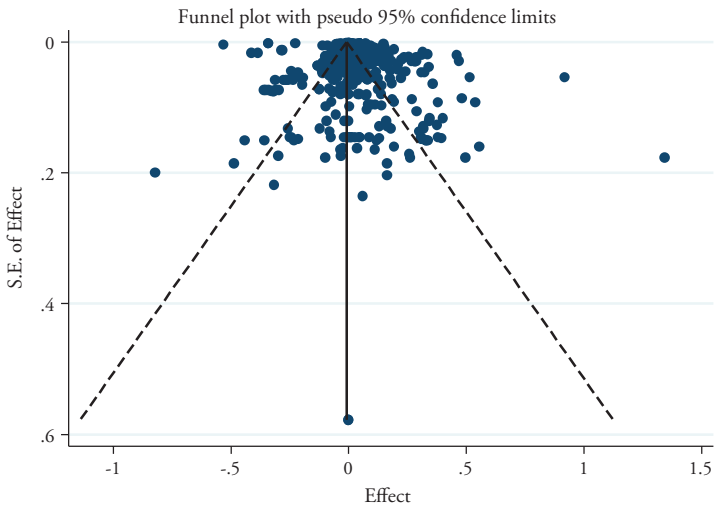

Table 3. Average Effect Sizes for the Effect of Government Grants on Private Donations in Nonprofit Organizations under Different Scenarios

Scenario	*CRVE Estimate*	*GEE Estimate*
Baseline	.0635	.0428
Baseline arts	.1555	.1388
Baseline education and research	.0746	.0535
Baseline environment	.0581	.0439
Baseline health care	.0852	.0852
Baseline human services	−.0013	−.0044
Baseline international development	.0756	.0494
Baseline public benefit	.0973	.0681
Baseline US nonprofits	.0076	.0194
Baseline age	.0211	.0001
Baseline size	.0558	.0395
Baseline longitudinal	.0648	.0440
Baseline lagged grants	.0645	.0566
Baseline endogeneity	.0251	.0045
Baseline peer-review	.0866	.0896
Highest-quality studies	.0071	.0098

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scenario, respectively. The average effect sizes are positive for most kinds of nonprofits using the results from the CRVE and the GEE models, with arts nonprofits demonstrating the largest effect. One exception is human service nonprofits with slight negative average effect sizes.

Overall, the correlation between government grants and private donations, no matter positive or negative, is fairly small. Moreover, the tenth through fifteenth rows show the average effect sizes from meta-regressions with each design component in turn under the baseline scenario. Again, all the average effect sizes are positive and relatively weak. In addition, the moderator analysis indicates that high-quality studies of interaction between government grants and private donations need to include organizational age as a control, rely on longitudinal data, correct for endogeneity, and appear in peer-reviewed outlets. Based on this finding, I estimated in the sixteenth row the average effect sizes of the highest-quality studies that meet all these four criteria. Similarly, the average effect sizes are positive and trivial.

In sum, the average effect sizes are positive under almost all scenarios using the results from the CRVE and the GEE models. The association between government grants and private contributions in nonprofits roughly ranges from approximately .01 in the highest-quality studies, to approximately .02 in the weighted average effect size analysis, to approximately .06 in meta-regression baseline analysis. Although these findings collectively indicate a positive correlation between government grants and private donations, the effect itself is actually negligible. Put another way, government grants may neither crowd-out nor crowd-in private donations.

Conclusion and Discussion

In recent decades, government collaboration with nonprofits for the delivery of public services and achievement of policy priorities has risen sharply. Although there is a subsector variation in the number of government grants and contracts obtained by nonprofits, the importance of government funding to the entire nonprofit sector is irreplaceable. Such a close funding relationship between the two sectors has attracted considerable scholarly attention from various perspectives. This study explored in particular the interplay between government grants and private donations in nonprofits. Although research on this question is flourishing, the findings are by far strikingly contradictory, causing difficulty in developing a coherent knowledge to guide both research and practice. The present study contributes to this body of literature by employing a meta-analysis to synthesize the competing findings. Based on a sample of sixty original studies with 637 effect sizes, the meta-analysis finds a positive correlation between government grants and private giving. However, the effect itself is too trivial to warrant either crowding-out or crowding-in explanation.

As mentioned previously, government grants to nonprofits could either displace or leverage private contributions through respective mechanisms, with the net effect dependent upon how donors and nonprofits themselves respond to government grants. The finding of the present study reminds us that these two funding sources may actually be independent or that the crowding-in and crowding-out forces may offset each other in the wrestling. Admittedly, the meta-analysis here could not determine the causal relationship between the two funding streams but only demonstrate the existence of a negligible positive association. However, the finding in this article does contribute to the literature by adding that the long-lasting scholarly debate on crowding-in and crowding-out might be overstated. Scholars may shift their attention from the tradeoff between the two funding streams to the capacity of attracting revenues from each source.

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The finding also has implications for nonprofit management. Given that most nonprofits operate under austerity conditions, developing an effective financial strategy to promote financial health and sustainability constitutes a pressing challenge. A common wisdom suggests diversifying revenue sources to minimize financial risks (Carroll and Stater 2009; Froelich 1999). However, this financial practice is often challenged by the uncertainty about whether different funding streams could be mutually exclusive or reinforcing. This study demonstrates that the association between government grants and private donations is minimal. Nonprofits thus could continue building on different combinations of public and private financial resources and seeking funding from these sources concurrently. Moreover, nonprofit leaders and managers should be more concerned about the capacity issue in competing for different types of funding, rather than the tradeoff among them. In particular, nonprofits should become more engaged in fundraising development and make fundraising capacity building a priority on organizational agenda (Greenfield 1999; Seiler, Aldrich, and Tempel 2010). It is very likely that nonprofits successfully drawing revenue from one funding source could develop more organizational capacity, which in turn enables them to more effectively compete for another source. In this way, more established organizations are better equipped to compete for different types of funding.

In addition to the estimation of average effect sizes, this study also explored the factors underlying effect size heterogeneity within and across studies. Through meta-regression, the study identified several moderators that contribute to high-quality research on the topic of interest. First, organizational age should be included as a control in that, because of information asymmetry, the impact of government is greater on younger organizations. Second, longitudinal data are superior to cross-sectional ones to track the dynamics over time. Third, government grants and private contribution could be jointly determined, and thus the endogeneity should be corrected. In sum, future studies should incorporate these components into their research and pay attention to subsector variation.

Moreover, during the literature review process, I discovered four potential future research areas. First, the impact of grants from different government levels has not been separately investigated sufficiently because most studies aggregate government grants from federal, state, and local levels. Steinberg (1987) called scholarly attention to the complexity of "joint crowdout," where federal funding is exogenous of private donations, while state and local funding is endogenous. Thus, the exploration of government grants on charitable donations interaction needs to consider the exogenous changes when the federal government transfers grants to state and local governments (Hughes et al. 2014; Knight 2002). Second, the impact of another type of government funding, service contracts, has not been carefully studied yet. Compared with grants, contracts are for well-defined services and have limited adaptation to a nonprofit's current activities. Thus, it is possible that the effect of contracts could be different. Third, philanthropic contributions to nonprofits come not only through monetary gifts but also through volunteer labor. A complete examination of the philanthropic consequences of government grants should address donations of both money and time (Callen 1994; Marcuello and Salas 2000). Examining money donations alone, as the original studies in our sample, may fail to capture the full spectrum of private contributions. Fourth, to fully detect the causality between government grants and private donations, researchers need to look at the interaction in another direction—the impact of private philanthropy on government support of nonprofits, for example, whether government takes a free ride on private philanthropy (Brooks 1999; Heutel 2014).

Overall, this study aimed to "build bodies of cumulative knowledge and provide more precise and robust guides for action" (Ringquist 2013, 9). Through a meta-analysis of sixty existing studies, it demonstrates that government grants have trivial impact on private donations to nonprofits. It informs that both researchers and practitioners should focus more on the capacity of competing for different types of funding, rather than the debate on crowding-in or crowding-out among them, in order to strengthen nonprofits' holistic financial performance and position.

Notes

1. Some studies on higher education organizations, such as Cheslock and Gianneschi (2008), used a combination of grants and appropriations established and mandated by law to measure government funding. I included these studies in the analysis.

2. When original studies reported only parameter estimates and indicated their statistical significance levels using asterisks, *t*-scores were estimated as the values at the symbol levels. When original studies reported only the parameters of interest that are not statistically significant, the effect sizes were set as 0. In these two situations, the effect sizes represent lower bound estimates (Ringquist 2013).

3. Some original studies did not specify the subsector(s) of their samples but implied a mix of different subsectors. In these cases, I coded the moderators on subsectors as 0s for these studies. However, this coding method may slightly bias the results.

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